

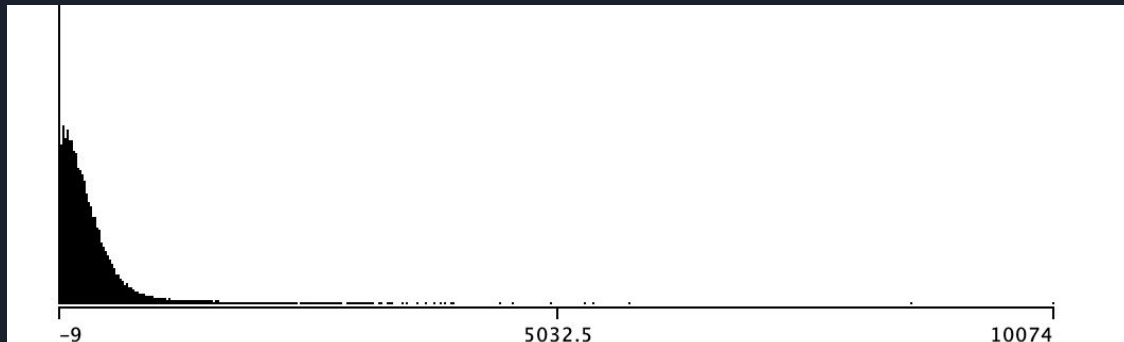


Predicting Free and Reduced Lunch Eligibility in US Public Schools

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Our Dataset

- Contains 101390 instances, dimensions 74
- administrative and fiscal data about all public schools, school districts, and state education agencies in the United States
- Contains special values : -1 or M indicates missing values and -2 or N indicates the data is not applicable. -9 indicates that the data does not meet NCES standards.
- Heavily skewed due to the right
- USDA income eligibility





Goal of Classification

- Aid in the allocation of financial and dietary resources to schools who need them most
- Predict the number of students that are eligible for free and reduced-price lunch.
- Class attribute (TOTFRL) which has the number of students on a reduced lunch program) is numeric and discrete.
- Split class data into two groups: low (below average) and high (above average)



Preprocessing

- removed all instances with missing class labels, leaving us with a total of 86,637 instances
 - We used Weka's remove SubsetByExpression with the expression "not ismissing(ATT#)" to remove attributes with missing class labels
- Removed unnecessary attributes: OBJECTID, NCESSCH, SURVYEAR, STABR, LEAID, ST_LEAID, LEA_NAME, SCH_NAME, LSTREET1, LSTREET2, PHONE, LATCOD, and LONCOD
- Removed all attributes that had missing values greater than 45% (PK, KG, G01-G13, AE, GSLO, GSHI, LZIP4, DIRECTCERT, and UG)
- Three special values: (-1 missing, -2 or N not applicable, -9 didn't meet surveyor requirements)
 - Removed with google sheets using replace tool
 - Treated as missing

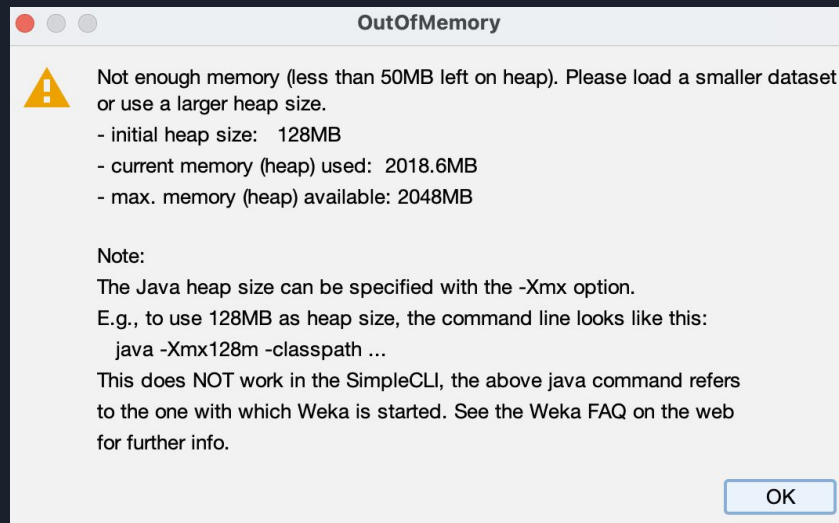


More Preprocessing

- Class Data (TOTFRL) discrete and numeric
- split the class data into two groups:
 - -low, meaning that the number of students eligible for free or reduced-price lunch is below average
 - high, meaning that the number of students eligible for free or reduced-price lunch is above average
- Used Weka's discretize filter with equal frequency binning

Pre-Training

- Memory Constraint because Dataset too large
- The nature of our class labels (high being above average and low being below average) lead to a perfect 50/50 split of instances among these two possible labels
- Used simple random sampling to produce a sample with a distribution of classes that is equal to the distribution of the original dataset





Train/Test split

- 70% of the data as a training set, 30% of the data as a test set.
- We split this test set in two, using 15% of our dataset as a validation set, and using another 15% of our dataset (the other half of the test set) as a final test set
- Used Weka's RemovePercentage filter to split the dataset into training and testing sets and to split the testing set into testing and validation sets.
- We ensured that the split datasets did not overlap by using each percentage once with and once without the InvertSelection field as true.

Attribute Selection - Correlation

- We used Weka's correlation attribute evaluator with the ranker search method. The results were as follows. We used a cutoff value of 0.2;
- the attributes we removed were: SCHOOL_TYPE_TEXT, HPALM, HP, WHALM, WH, HPALF, WHALF, CHARTER_TEXT, AS_, ASALM, ASALF, ULOCALE, STUTERATIO, SY_STATUS_TEXT, LSTATE, STATUS, AMALM, AM, AMALF, VIRTUAL, SCHOOL_LEVEL, LZIP, NMCNTY, and LCITY.
- The attributes with a correlation greater than 0.2 were: MEMBER, TOTAL, TOTMENROL, TOTFENROL, FTE, HI, HIALM, HIALF, BLALM, BL, BLALF, TR, TRALF, and TRALM.

The screenshot shows the Weka software interface with the 'Attribute Selection' tab active. The 'Attribute Evaluator' is set to 'CorrelationAttributeEval' and the 'Search Method' is 'Ranker'. The 'Attribute Selection Mode' is 'Use full training set'. The 'Attribute selection output' pane displays the results of the selection process, showing a list of ranked attributes and the selected attributes.

Attribute selection output

Attribute ranking.

Attribute Evaluator (supervised, Class (nominal): 39 TOTFRL):

Correlation Ranking Filter

Ranked attributes:

Rank	Attribute
0.4589	15 MEMBER
0.4587	14 TOTAL
0.4574	12 TOTMENROL
0.4521	13 TOTFENROL
0.4444	16 FTE
0.4393	32 HI
0.4393	30 HIALM
0.4365	31 HIALF
0.3964	24 BLALM
0.3946	26 BL
0.3898	25 BLALF
0.2674	35 TR
0.2653	34 TRALF
0.2616	33 TRALM
0.1834	8 SCHOOL_TYPE_TEXT
0.0944	27 HPALM
0.0941	29 HP
0.0936	36 WHALM
0.0929	38 WH
0.0921	28 HPALF
0.0918	37 WHALF
0.0822	4 CHARTER_TEXT
0.0738	23 AS_
0.0738	21 ASALM
0.0734	22 ASALF
0.0709	10 ULOCALE
0.0633	17 STUTERATIO
0.058	9 SY_STATUS_TEXT
0.057	2 LSTATE
0.0562	7 STATUS
0.0487	18 AMALM
0.046	28 AM
0.0428	19 AMALF
0.038	5 VIRTUAL
0.0346	6 SCHOOL_LEVEL
0.023	3 LZIP
0.02	11 NMCNTY
0.0151	1 LCITY

Selected attributes: 15, 14, 12, 13, 16, 32, 30, 31, 24, 26, 25, 35, 34, 33, 8, 27, 29, 36, 38, 28, 37, 4, 23, 21, 22, 10, 17, 9, 2, 7, 18, 20, 19, 5, 6, 3, 11, 1 : 38

Info Gain

- We used Weka's info gain attribute evaluator with the ranker search method.
- We used a cutoff value of 0.2
- The attributes we removed were: LSTATE, LZIP, ULOCALE, TR, STUTERATIO, TRALF, TRALM, AS_, HP, ASALM, ASALF, HPALM, HPALF, AM, AMALM, SCHOOL_TYPE_TEXT, WH, WHALM, WHALF, AMALF, SCHOOL_LEVEL, CHARTER_TEXT, VIRTUAL, SY_STATUS_TEXT, and STATUS.
- The attributes with an info gain greater than 0.2 were: LCITY, MEMBER, TOTAL, TOTMENROL, TOTFENROL, NMCNTY, HI, HIALF, HIALM, FTE, BL, BLALF, and BLALM.
- Infogain is a measure of how well a variable reduces the entropy of another variable, calculated by the equation

$$\text{InfoGain}(\text{Class}, \text{Attribute}) = H(\text{Class}) - H(\text{Class} \mid \text{Attribute}).$$

The screenshot shows the Weka software interface, specifically the 'Attribute Evaluator' window. The 'InfoGainAttributeEval' method is selected. The 'Search Method' is set to 'Ranker' with a cutoff of -1.7976931348623157E308. The 'Attribute Selection Mode' is set to 'Use full training set'. The 'Ranked attributes' list is displayed, showing attributes ranked by their info gain. The top attributes are LCITY, MEMBER, TOTAL, TOTMENROL, TOTFENROL, NMCNTY, HI, HIALF, HIALM, FTE, BL, BLALF, and BLALM. The 'Selected attributes' list at the bottom shows the attributes selected for the model.

Attribute Evaluator

Choose **InfoGainAttributeEval**

Search Method

Choose **Ranker** -T -1.7976931348623157E308 -N -1

Attribute Selection Mode

☒ Use full training set

☐ Cross-validation

Folds: 10

Seed: 1

No class

Start Stop

Result list (right-click for options)

00:18:50 - Ranker + CorrelationAttributeEval

00:30:14 - Ranker + InfoGainAttributeEval

Attribute selection output

Attribute Evaluator (supervised, Class (nominal): 39 TOTFRL):
Information Gain Ranking Filter

Ranked attributes:

Info Gain	Attribute
0.64694	1 LCITY
0.3468	15 MEMBER
0.34656	14 TOTAL
0.33231	12 TOTMENROL
0.33142	13 TOTFENROL
0.38848	11 NMCNTY
0.27594	32 HI
0.27183	31 HIALF
0.26959	38 HIALM
0.253	16 FTE
0.23469	26 BL
0.23133	25 BLALF
0.23119	24 BLALM
0.11633	2 LSTATE
0.11268	3 LZIP
0.47387	10 ULOCALE
0.46987	35 TR
0.46838	17 STUTERATIO
0.4667	34 TRALF
0.46428	33 TRALM
0.44923	23 AS_
0.44824	29 HP
0.44542	21 ASALM
0.44817	22 ASALF
0.43878	27 HPALM
0.43525	28 HPALF
0.43322	28 WH
0.4327	18 AMALM
0.4313	8 SCHOOL_TYPE_TEXT
0.42884	38 WH
0.42744	36 WHALM
0.42523	37 WHALF
0.42874	19 AMALF
0.41643	6 SCHOOL_LEVEL
0.41337	4 CHARTER_TEXT
0.40791	5 VIRTUAL
0.40283	9 SY_STATUS_TEXT
0.40262	7 STATUS

Selected attributes: 1,15,14,12,13,11,32,31,30,16,26,25,24,2,3,10,35,17,34,33,23,29,21,22,27,28,20,18,8,38,36,37,19,6,4,5,9,7 : 38

Status OK

Log x 0

Symmetrical Uncertainty

- used a cutoff value of 0.1
- the attributes removed were: NMCNTY, LZIP, STUTERATIO, SCHOOL_TYPE_TEXT, TRALF, TRALM, TR, LSTATE, HP, HPALM, AS_, ASALM, ULOCALE, ASALF, HPALF, AM, AMALM, WH, WHALM, AMALF, AMALF, WHALF, CHARTER_TEXT, SCHOOL_LEVEL, VIRTUAL, SY_STATUS_TEXT, and STATUS.
- The attributes with a symmetrical uncertainty value greater than 0.1 were: MEMBER, TOTAL, TOTFENROL, TOTMENROL, HIALF, HI, HIALM, FTE, BL, BLALF, BLALM, and LCITY.

Preprocess Classify Cluster Associate **Select attributes** Visualize

Attribute Evaluator
Choose: **SymmetricalUncertAttributeEval**

Search Method
Choose: **Ranker -T -1.7976931348623157E308 -N -1**

Attribute Selection Mode
☒ Use full training set
☐ Cross-validation Folds: 10 Seed: 1

No class

Start Stop

Result list (right-click for options)
02:14:44 - BestFirst + WrapperSubsetEval
02:15:02 - Ranker + SymmetricalUncertAttributeEval

Attribute selection output
Attribute Evaluator (supervised, Class (nominal): 39 TOTFRL):
Symmetrical uncertainty Ranking Filter

Ranked attributes:

0.1968	15	MEMBER
0.1967	14	TOTAL
0.1866	13	TOTFENROL
0.1858	12	TOTMENROL
0.1489	31	HIALF
0.1471	32	HI
0.1443	30	HIALM
0.1403	16	FTE
0.1273	26	BL
0.1225	25	BLALF
0.1218	24	BLALM
0.1052	1	LCITY
0.0607	11	NMCNTY
0.0515	3	LZIP
0.0403	17	STUTERATIO
0.0401	8	SCHOOL_TYPE_TEXT
0.0401	34	TRALF
0.0401	33	TRALM
0.0395	35	TR
0.0384	2	LSTATE
0.0378	29	HP
0.0379	27	HPALM
0.0357	23	AS_
0.0351	21	ASALM
0.0351	18	ULOCALE
0.0350	22	ASALF
0.0331	28	HPALF
0.0263	20	AM
0.0227	18	AMALM
0.0212	38	WH
0.0206	36	WHALM
0.0201	19	WHALF
0.0201	37	WHALF
0.0179	4	CHARTER_TEXT
0.0121	6	SCHOOL_LEVEL
0.0066	5	VIRTUAL
0.0034	9	SY_STATUS_TEXT
0.0049	7	STATUS

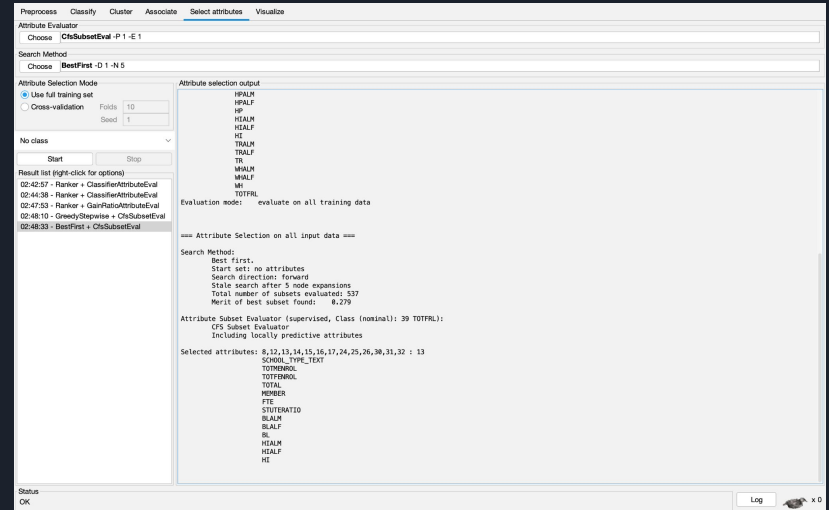
Selected attributes: 15, 14, 13, 12, 31, 32, 30, 16, 26, 25, 24, 1, 11, 3, 17, 8, 34, 33, 35, 2, 29, 27, 23, 21, 18, 22, 28, 20, 18, 38, 36, 19, 37, 4, 6, 5, 9, 7, 38

Status
OK

Log x 0

Cfs Subset

- Attributes kept: SCHOOL_TYPE_TEXT, TOTMENROL, TOTFENROL, TOTAL, MEMBER, FTE, STUTERATIO, BLALM, BLALF, BL, HIALM, HIALF, and HI.
- CfsSubsetAttributeEval evaluates attributes by how correlated they are with the class, while also attempting to avoid selecting attributes highly correlated with each other





Intuition

- Using our intuition. We considered how heavily we thought attributes might be related to the class.
- Picked the following attributes: LSTATE, LZIP, SCHOOL_LEVEL, ULOCALE, NMCITY, TOTAL, STUTERATIO, AM, AS_, BL, HP, HI, TR, and WH.
- All other attributes were removed.



Classification

- NaiveBayes
 - Probability
 - Assumes all attributes are independent
- J48
 - Weka C4.5
 - Information entropy
 - For each attribute a , find normalised info gain ratio
 - Create a node that splits based on attribute with highest info gain
 - Recur
- 1R
 - Single rule
- Random Tree
 - Decision tree with k randomly chosen attributes at each node

Classification - CorrelationAttributeEval

- Naive Bayes : Training accuracy was ~81%, validation accuracy was ~79%
- J48: Training accuracy was ~88%, validation accuracy was ~84%
- 1R: Training accuracy was ~79%, validation accuracy was ~74%
- Random Tree: Training accuracy was ~100%, validation accuracy was ~80%

```
=== Summary ===
Correctly Classified Instances      1049      80.7544 %
Incorrectly Classified Instances    250      19.2456 %
Kappa statistic                    0.6136
Mean absolute error                0.189
Root mean squared error            0.422
Relative absolute error            37.7942 %
Root relative squared error        84.4036 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
Weighted Avg.   0.698   0.079   0.895   0.898   0.779   0.630   0.916   0.916   '(-inf-217.5]
Weighted Avg.   0.888   0.196   0.623   0.888   0.805   0.630   0.916   0.914   '(217.5-inf)'

=== Confusion Matrix ===
  a  b  <-- classified as
608 52 | a = '(-inf-217.5]'
198 441 | b = '(217.5-inf)'
```

```
=== Summary ===
Correctly Classified Instances      961      73.98 %
Incorrectly Classified Instances    338      26.02 %
Kappa statistic                    0.4801
Mean absolute error                0.2602
Root mean squared error            0.5101
Relative absolute error            52.0435 %
Root relative squared error        102.0256 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
Weighted Avg.   0.708   0.227   0.763   0.708   0.734   0.481   0.740   0.689   '(-inf-217.5]'
Weighted Avg.   0.773   0.292   0.719   0.773   0.745   0.481   0.740   0.668   '(217.5-inf)'

=== Confusion Matrix ===
  a  b  <-- classified as
467 193 | a = '(-inf-217.5]'
145 494 | b = '(217.5-inf)'
```

```
=== Summary ===
Correctly Classified Instances      1097      84.4496 %
Incorrectly Classified Instances    202      15.5504 %
Kappa statistic                    0.6884
Mean absolute error                0.2018
Root mean squared error            0.3419
Relative absolute error            40.3561 %
Root relative squared error        68.3787 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
Weighted Avg.   0.897   0.210   0.815   0.897   0.854   0.692   0.907   0.899   '(-inf-217.5]'
Weighted Avg.   0.790   0.103   0.881   0.790   0.833   0.692   0.907   0.874   '(217.5-inf)'

=== Confusion Matrix ===
  a  b  <-- classified as
592 68 | a = '(-inf-217.5]'
134 505 | b = '(217.5-inf)'
```

```
=== Summary ===
Correctly Classified Instances      1035      79.6767 %
Incorrectly Classified Instances    264      20.3233 %
Kappa statistic                    0.5937
Mean absolute error                0.2832
Root mean squared error            0.4508
Relative absolute error            40.6494 %
Root relative squared error        90.168 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
Weighted Avg.   0.780   0.186   0.812   0.780   0.796   0.594   0.797   0.745   '(-inf-217.5]'
Weighted Avg.   0.814   0.220   0.782   0.814   0.798   0.594   0.797   0.728   '(217.5-inf)'

=== Confusion Matrix ===
  a  b  <-- classified as
515 145 | a = '(-inf-217.5]'
119 520 | b = '(217.5-inf)'
```

Classification - Info Gain

- NaiveBayes: Training accuracy was ~82%, validation accuracy was ~80%
- J48: Training accuracy was ~82%, validation accuracy was ~82%
- 1R: Training accuracy was ~86%, validation accuracy was ~63%
- Random Tree: Training accuracy was ~100%, validation accuracy was ~76%

=== Summary ===

Correctly Classified Instances	1058	81.4473 %
Incorrectly Classified Instances	241	18.5527 %
Kappa statistic	0.6274	
Mean absolute error	0.1845	
Root mean squared error	0.4157	
Relative absolute error	36.8963 %	
Root relative squared error	83.1537 %	
Total Number of Instances	1299	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.932	0.307	0.758	0.932	0.836	0.645	0.918	0.916	'(-inf-217.5)'
	0.693	0.068	0.908	0.693	0.786	0.645	0.918	0.919	'(217.5-inf)'
Weighted Avg.	0.814	0.189	0.832	0.814	0.812	0.645	0.918	0.917	

=== Confusion Matrix ===

a	b	<-- classified as
615	45	a = '(-inf-217.5)'
196	443	b = '(217.5-inf)'

=== Summary ===

Correctly Classified Instances	810	62.3557 %
Incorrectly Classified Instances	489	37.6443 %
Kappa statistic	0.2433	
Mean absolute error	0.3764	
Root mean squared error	0.6135	
Relative absolute error	75.2937 %	
Root relative squared error	122.7171 %	
Total Number of Instances	1299	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.776	0.534	0.600	0.776	0.677	0.255	0.621	0.580	'(-inf-217.5)'
	0.466	0.224	0.668	0.466	0.549	0.255	0.621	0.574	'(217.5-inf)'
Weighted Avg.	0.624	0.381	0.634	0.624	0.614	0.255	0.621	0.577	

=== Confusion Matrix ===

a	b	<-- classified as
512	148	a = '(-inf-217.5)'
341	298	b = '(217.5-inf)'

=== Summary ===

Correctly Classified Instances	1076	82.8329 %
Incorrectly Classified Instances	223	17.1671 %
Kappa statistic	0.6552	
Mean absolute error	0.2378	
Root mean squared error	0.3459	
Relative absolute error	47.5618 %	
Root relative squared error	69.1805 %	
Total Number of Instances	1299	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.948	0.296	0.768	0.948	0.849	0.675	0.896	0.870	'(-inf-217.5)'
	0.704	0.052	0.930	0.704	0.801	0.675	0.896	0.859	'(217.5-inf)'
Weighted Avg.	0.828	0.176	0.848	0.828	0.826	0.675	0.896	0.864	

=== Confusion Matrix ===

a	b	<-- classified as
626	34	a = '(-inf-217.5)'
189	450	b = '(217.5-inf)'

=== Summary ===

Correctly Classified Instances	989	76.1355 %
Incorrectly Classified Instances	310	23.8645 %
Kappa statistic	0.5242	
Mean absolute error	0.2586	
Root mean squared error	0.4368	
Relative absolute error	51.717 %	
Root relative squared error	87.3633 %	
Total Number of Instances	1299	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.664	0.138	0.833	0.664	0.739	0.536	0.814	0.781	'(-inf-217.5)'
	0.862	0.336	0.713	0.862	0.780	0.536	0.814	0.759	'(217.5-inf)'
Weighted Avg.	0.761	0.235	0.774	0.761	0.759	0.536	0.814	0.770	

=== Confusion Matrix ===

a	b	<-- classified as
438	222	a = '(-inf-217.5)'
88	551	b = '(217.5-inf)'

Classification - SymmetricalUncertAttributeEval

- NaiveBayes: Training accuracy was ~81%, validation accuracy was ~80%
- J48: Training accuracy was ~83%, validation accuracy was ~82%
- 1R: Training accuracy was ~86%, validation accuracy was ~64%
- Random Tree: Training accuracy was ~100%, validation accuracy was ~75%

```
=== Summary ===
Correctly Classified Instances      1061      81.6782 %
Incorrectly Classified Instances    238      18.3218 %
Kappa statistic                    0.6321
Mean absolute error                0.1842
Root mean squared error            0.4155
Relative absolute error             36.8419 %
Root relative squared error        83.1044 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
          TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
          0.935   0.305   0.760     0.935   0.838     0.650   0.919   0.919   '(-inf-217.5]'
          0.695   0.065   0.912     0.695   0.789     0.650   0.919   0.919   '(217.5-inf)'
Weighted Avg.    0.817   0.187   0.835     0.817   0.814     0.650   0.919   0.919

=== Confusion Matrix ===
  a  b  <-- classified as
617 43 | a = '(-inf-217.5]'
195 444 | b = '(217.5-inf)'
```

```
=== Summary ===
Correctly Classified Instances      1079      83.0639 %
Incorrectly Classified Instances    220      16.9361 %
Kappa statistic                    0.6598
Mean absolute error                0.2326
Root mean squared error            0.3433
Relative absolute error             46.5263 %
Root relative squared error        68.6575 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
          TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
          0.956   0.299   0.768     0.956   0.852     0.682   0.899   0.870   '(-inf-217.5]'
          0.701   0.044   0.939     0.701   0.803     0.682   0.899   0.867   '(217.5-inf)'
Weighted Avg.    0.831   0.173   0.852     0.831   0.828     0.682   0.899   0.868

=== Confusion Matrix ===
  a  b  <-- classified as
631 29 | a = '(-inf-217.5]'
191 448 | b = '(217.5-inf)'
```

```
=== Summary ===
Correctly Classified Instances      810      62.3557 %
Incorrectly Classified Instances    489      37.6443 %
Kappa statistic                    0.2433
Mean absolute error                0.3764
Root mean squared error            0.6135
Relative absolute error             75.2937 %
Root relative squared error        122.7171 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
          TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
          0.776   0.534   0.600     0.776   0.677     0.255   0.621   0.580   '(-inf-217.5]'
          0.466   0.224   0.668     0.466   0.549     0.255   0.621   0.574   '(217.5-inf)'
Weighted Avg.    0.624   0.381   0.634     0.624   0.614     0.255   0.621   0.577

=== Confusion Matrix ===
  a  b  <-- classified as
512 148 | a = '(-inf-217.5]'
341 298 | b = '(217.5-inf)'
```

```
=== Summary ===
Correctly Classified Instances      959      73.826 %
Incorrectly Classified Instances    340      26.174 %
Kappa statistic                    0.4757
Mean absolute error                0.3452
Root mean squared error            0.5029
Relative absolute error             69.0425 %
Root relative squared error        100.5791 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
          TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
          0.782   0.307   0.725     0.782   0.752     0.477   0.733   0.669   '(-inf-217.5]'
          0.693   0.218   0.755     0.693   0.723     0.477   0.733   0.692   '(217.5-inf)'
Weighted Avg.    0.738   0.263   0.739     0.738   0.738     0.477   0.733   0.680

=== Confusion Matrix ===
  a  b  <-- classified as
516 144 | a = '(-inf-217.5]'
196 443 | b = '(217.5-inf)'
```


Classification - CfsSubsetAttributeEval

- NaiveBayes: Training accuracy was ~81%, validation accuracy was ~80%
- J48: Training accuracy was 88%, validation accuracy was ~84%
- 1R: Training accuracy was ~79%, validation accuracy was ~74%
- Random Tree: Training accuracy was ~100%, validation accuracy was ~80%

```
=== Summary ===
Correctly Classified Instances      1058      81.4473 %
Incorrectly Classified Instances    241      18.5527 %
Kappa statistic                    0.6274
Mean absolute error                0.185
Root mean squared error            0.4168
Relative absolute error             36.9942 %
Root relative squared error         83.3643 %
Total Number of Instances          1299

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
Weighted Avg.   0.933   0.388   0.933      0.933   0.836      0.646   0.921   0.923   '(-inf-217.5]
               0.692   0.067   0.909      0.692   0.786      0.646   0.921   0.928   '(217.5-inf)'

=== Confusion Matrix ===
  a  b  <-- classified as
616 44 | a = '(-inf-217.5]'
197 442 | b = '(217.5-inf)'
```

```
=== Summary ===
Correctly Classified Instances      961      73.98 %
Incorrectly Classified Instances    338      26.02 %
Kappa statistic                    0.4901
Mean absolute error                0.2602
Root mean squared error            0.5101
Relative absolute error             52.0435 %
Root relative squared error         102.0256 %
Total Number of Instances          1299

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
Weighted Avg.   0.708   0.227   0.763      0.708   0.734      0.481   0.740   0.689   '(-inf-217.5]
               0.773   0.292   0.719      0.773   0.745      0.481   0.740   0.668   '(217.5-inf)'

=== Confusion Matrix ===
  a  b  <-- classified as
467 193 | a = '(-inf-217.5]'
145 494 | b = '(217.5-inf)'
```

```
=== Summary ===
Correctly Classified Instances      1092      84.0647 %
Incorrectly Classified Instances    207      15.9353 %
Kappa statistic                    0.6808
Mean absolute error                0.2002
Root mean squared error            0.3307
Relative absolute error             40.0443 %
Root relative squared error         67.7479 %
Total Number of Instances          1299

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
Weighted Avg.   0.883   0.203   0.818      0.883   0.849      0.683   0.911   0.901   '(-inf-217.5]
               0.797   0.117   0.869      0.797   0.831      0.683   0.911   0.881   '(217.5-inf)'

=== Confusion Matrix ===
  a  b  <-- classified as
583 77 | a = '(-inf-217.5]'
130 509 | b = '(217.5-inf)'
```

```
=== Summary ===
Correctly Classified Instances      1029      79.2148 %
Incorrectly Classified Instances    270      20.7852 %
Kappa statistic                    0.5842
Mean absolute error                0.2079
Root mean squared error            0.4559
Relative absolute error             41.5732 %
Root relative squared error         91.1869 %
Total Number of Instances          1299

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
Weighted Avg.   0.797   0.213   0.795      0.797   0.796      0.584   0.792   0.736   '(-inf-217.5]
               0.787   0.203   0.790      0.787   0.788      0.584   0.792   0.726   '(217.5-inf)'

=== Confusion Matrix ===
  a  b  <-- classified as
526 134 | a = '(-inf-217.5]'
136 503 | b = '(217.5-inf)'
```

Classification - Intuition

- NaiveBayes: Training accuracy was ~80%, validation accuracy was ~80%
- J48: Training accuracy was 84%, validation accuracy was ~84%
- 1R: Training accuracy was ~79%, validation accuracy was ~74%
- Random Tree: Training accuracy was ~100%, validation accuracy was ~75%

```
=== Summary ===
Correctly Classified Instances      1044      80.3695 %
Incorrectly Classified Instances    255      19.6305 %
Kappa statistic                    0.6059
Mean absolute error                0.199
Root mean squared error            0.4195
Relative absolute error            39.807 %
Root relative squared error        83.8957 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
               0.915    0.311    0.752     0.915    0.826     0.621    0.905    0.909    '(-inf-217.5]'
               0.689    0.085    0.887     0.689    0.775     0.621    0.905    0.898    '(217.5-inf)'
Weighted Avg.   0.804    0.200    0.819     0.804    0.801     0.621    0.905    0.903

=== Confusion Matrix ===
      a  b  <-- classified as
604 56 | a = '(-inf-217.5]'
199 440 | b = '(217.5-inf)'
```

```
=== Summary ===
Correctly Classified Instances      961      73.98 %
Incorrectly Classified Instances    338      26.02 %
Kappa statistic                    0.4801
Mean absolute error                0.2602
Root mean squared error            0.5101
Relative absolute error            52.0435 %
Root relative squared error        102.0256 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
               0.708    0.227    0.763     0.708    0.734     0.481    0.740    0.689    '(-inf-217.5]'
               0.773    0.292    0.719     0.773    0.745     0.481    0.740    0.668    '(217.5-inf)'
Weighted Avg.   0.740    0.259    0.741     0.740    0.740     0.481    0.740    0.678

=== Confusion Matrix ===
      a  b  <-- classified as
467 193 | a = '(-inf-217.5]'
145 494 | b = '(217.5-inf)'
```

```
=== Summary ===
Correctly Classified Instances      1078      82.9869 %
Incorrectly Classified Instances    221      17.0131 %
Kappa statistic                    0.6587
Mean absolute error                0.2271
Root mean squared error            0.3424
Relative absolute error            45.424 %
Root relative squared error        68.481 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
               0.918    0.261    0.784     0.918    0.846     0.669    0.905    0.883    '(-inf-217.5]'
               0.739    0.082    0.897     0.739    0.810     0.669    0.905    0.875    '(217.5-inf)'
Weighted Avg.   0.830    0.173    0.840     0.830    0.828     0.669    0.905    0.879

=== Confusion Matrix ===
      a  b  <-- classified as
606 54 | a = '(-inf-217.5]'
167 472 | b = '(217.5-inf)'
```

```
=== Summary ===
Correctly Classified Instances      957      73.6721 %
Incorrectly Classified Instances    342      26.3279 %
Kappa statistic                    0.473
Mean absolute error                0.273
Root mean squared error            0.4998
Relative absolute error            54.688 %
Root relative squared error        99.9706 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
               0.761    0.288    0.732     0.761    0.746     0.473    0.746    0.694    '(-inf-217.5]'
               0.712    0.239    0.742     0.712    0.727     0.473    0.746    0.685    '(217.5-inf)'
Weighted Avg.   0.737    0.264    0.737     0.737    0.737     0.473    0.746    0.689

=== Confusion Matrix ===
      a  b  <-- classified as
502 158 | a = '(-inf-217.5]'
184 455 | b = '(217.5-inf)'
```



Analysis

- Random Trees can be seen overfitting (100% on training data, then worse from other models on testing)
- Best Model in terms of accuracy on testing and training was J48 in Intuition, CfsSubsetAttributeEval, and CorrelationAttributeEval and 1R in SymmetricalUncertAttributeEval
- J48 better perform than 1R
- Best Model in terms of ROC area is CfsSubsetAttributeEval with NaiveBayes with value of 0.921
- Best Model in terms of F-score is J48 in both CorrelationAttributeEval and CfsSubsetAttributeEval with value of 0.84



Conclusion

Issues:

- Dataset was originally too large to just run classifying
- Lots of missing values in our attributes, reducing the amount of data available
- Unable to run PCA

Areas for improvement:

- Running more classification models to get a better understanding of our attributes
- Training accuracy can be improved by increasing representation of all of our different attributes
- Discretize data into more categories